Assessment 2: Tutorial Submission 2 - Written Report

Submitted By: u3335040

Introduction

The report describes the issues faced by a Sydney-based design firm, which has recently debuted their new product range of outdoor pots. However, their inventory processes are still manual, leading to stock inaccuracies, and lack of synchronisation with the other software platforms. As a result, the organisation has no single 'source of truth' for its new product line's business data.

Main Body

Problem Description

The Sydney-based firm, **Root and Branch** (not their real name) has released its new product range of outdoor pots and planters. Unfortunately, the current inventory processes are manual, with spreadsheets still being used for stock counts, this has led to immediate inaccuracy issues such as 'phantom stock' (inventory physically missing but marked in stock), lack of real-time data and no synchronisation with other platforms, especially their accounting system, which prevents accurate financial reporting on the new product line.

This case explores the classic social-technical problem of **siloed data and fragmented systems** within an organisation, which occurs when a 'patchwork' of legacy systems, prevents an organization from achieving a unified view of its operations (Davenport 1998).

Stakeholders and Skills Mapping

Stakeholder Map

Three Stakeholder Groups, the Warehouse Team, Finance Department and Executive-level Management (shortened to **Warehouse/Finance/Executives**), have been identified for this case and are elaborated on below:

Group	Role	Interest	Main Concern
Warehouse	Data Entry and order fulfilment	Physical safety and process efficiency	Bottleneck during peak orders, Workplace accidents

Finance	Financial reporting	Accurate inventory valuation and overhead visibility	Lack of visibility due to siloed warehouse data resulting in financial inaccuracies
Executives	Operational and Strategic Oversight	Return-on-Investment, Brand Reputation and Legal Compliance	Cost overruns or a client data breach causing reputational damage

SFIA Skills Map (SFIA Foundation 2024)

Group	SFIA Skill (Code) / Level	Reason
Warehouse	Asset Management (ASMG) / L3	Necessary to manage order fulfilment while enforcing safety guardrails
Finance	Financial Analysis (FIAN) / L4	Ensuring the proposed system does not just track inventory physically in the warehouse, but also as a financial asset
Executives	Governance (GOVN) / L5	Govern the overall investment and provide the framework to ensure the system meets legal and ethical obligations

*The latest version of the SFIA v9 no longer has Inventory Management or Logistics related skills, Asset Management is the current closest equivalent

Critical involvement and Relationship to Solutions

Group	How they relate to the solution	Why their involvement is Critical?
Warehouse	Enforce Guardrail R1 (Warehouse Safety)*	Ensure data entered is reconciled with the warehouse shelf.
Finance	Ensure integration between warehouse solution and accounting system	Ensure new warehouse operations translate to accurate inventory valuation and financial reporting
Executives	Provide the strategic direction and enforce the R2 (Data Privacy) and R3 (Ethical Supplier) Guardrails*	Executive-level management determines the resource allocation and risk-tolerance for the new inventory solution

*See appendix for a description of the Guardrails

Proposed Solutions and Rationale

Solution 1: Centralized Inventory Management System / Hub
This solution involves an inventory management system that act as the 'source-of-truth' for inventory data that will sync with the other subsystems in the company (the accounting, online store, project management system etc) via API.

Solution 2: Comprehensive ERP Rollout
This solution replaces **all** the fragmented systems in the company with a single Enterprise Resource Planning (ERP) Suite (i.e. comprehensive software platform that integrates all core business processes; inventory, accounting etc).

Areas of Interest:	Solution 1: Inventory Hub	Solution 2: ERP Rollout
Technical Feasibility and Implementation	High. Off-the-shelf software solution designed for rapid deployment.	Medium to Low: Heavy technical requirements, large-scale data migration.
Socio-Technical Alignment	Solves the data fragmentation problem with a 'low user friction' solution.	While the superior 'technical' solution, may not pass the 'social' test due to high complexity and rigidity.
Risks and Mitigation	Require synchronization with other software systems via API (i.e. vulnerable to a breakdown or lag) can be mitigated via a status dashboard.	High risk of scope creep and cost overruns, mitigated by extensive change management and a multi-phase deployment.
Resources	Moderate cost and short timeframe (4-6 weeks for a medium sized company).	High cost and long timeframe (6–10 months for a medium sized company).

Stakeholder Impact and Acceptance

Stakeholders	Solution 1: Inventory Hub	Solution 2: ERP Rollout
Warehouse	Reduces administrative load by reducing manual reconciliation and processes / High Acceptance	While impact is the same as the previous solution, there may be a very high learning curve due to systems complexity / Low acceptance

Finance	Solves the ‘phantom stock’ issue and improves inventory valuation via API / High Acceptance	All data is ‘native’ since it’s a single system, but migration is risky / Medium acceptance
Executives	Rapid ROI and modest initial investment / High Acceptance	Requires significant capital expenditure and a long implementation period / Medium-to-Low acceptance

Recommendation

RECOMMENDATION: SOLUTION 1 (INVENTORY HUB)

Solution 1 is recommended as the more viable solution for the company, it addresses the socio-technical problem of fragmented data systems with an API-driven single ‘source-of-truth' at lower cost and deployment time (while also enforcing the chosen Guardrails). It also provides a scalable digital base for future transition to a full ERP as maturity evolves.

Feedback and Revisions

I will briefly discuss two feedback blocks I received.

The first was in the **Week 3 Tutorial (Stakeholders and Skills)** and I was asked for more clarification on the ‘Impact and Influence’ matrix in my notes, as it lacked the detail required to predict how stakeholder groups might resist or champion the change.

The second was in **Week 8 (Report Builder Studio)** and it was to work more on my evidence and gap list (relating to the missing ‘Options’ and ‘Decision’ sections) as it did not have enough comparative rigor to meet report requirements.

I agreed with both feedback blocks, and a condensed version of these improvements is in the ‘Stakeholders’ and ‘Solutions’ section of this report.

Appendix

Reference List

Davenport, TH 1998, 'Putting the enterprise into the enterprise system', *Harvard Business Review*, vol. 76, no. 4, pp. 121–131.

SFIA Foundation 2024, SFIA 9: The global skills and competency framework for the digital world, *SFIA Foundation*, viewed 12 April 2026, https://sfia-online.org/en/sfia-9/skills/all-skills-a-z.

Guardrails

Guardrail	Description
R1 (Warehouse Safety)	No manually lifting objects heavier than 25KG
R2 (Protect Client Data)	Take steps to protect customer's personal information
R3 (Ethical Suppliers)	Validate manufacturers ethical labour policies

Feedback Blocks

CHANGE BLOCK:

Peer name and ID: Kushalveer Singh Chahal (u3323455)
One suggestion (clarity/distinctness): Just clarify a couple points on the impact and influence
One trade-off/risk to add: The rest is pretty good
Signature: KSC **Date:** 13 / 3 / 26

CHANGE BLOCK:

Peer name and ID: Krystle Anne Orpia (u3318521)
One suggestion (clarity/distinctness): Complete the Part B: evidence map and Part C: Gap list + week 9 plan.
One trade-off/risk to add: Details from Part A are so detailed and it's good for your blueprint project.
Signature: KAO **Date:** 10 / 4 / 26

GenAI Acknowledgement Statement

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- **GenAI service name and version**: Microsoft Copilot
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